are due to differences in the structure and composition of the cell wall. The difference between Gram Positive and Gram negative bacteria is given in Table 1.6.

**What are Magnetosomes ?**

Intracellular chains of 40-50 magnetite ($Fe_3O_4$) particles are found in bacterium *Aquaspirillum magnetotacticum*. and it help the bacterium to locate nutrient rich sediments.

### 1.4.5 Life processes in Bacteria

#### *Respiration*

Two types of respiration are found in Bacteria. They are 1. Aerobic respiration 2. Anaerobic respiration.

**1. Aerobic respiration**

These bacteria require oxygen as terminal acceptor and will not grow under anaerobic conditions. (i.e. in the absence of $O_2$) **Example:** *Streptococcus.*

**Obligate aerobes**

Some *Micrococcus* **species** are obligate aerobes (i.e. they must have oxygen to survive).

**2. Anaerobic respiration**

These bacteria do not use oxygen for growth and metabolism but obtain their energy from fermentation reactions.**Example:** *Clostridium.*

**Facultative anaerobes**

There are bacteria that can grow either using oxygen as a terminal electron acceptor or anaerobically using fermentation reaction to obtain energy. When a facultative anaerobe such as *E. coli* is present at a site of infection like an abdominal abscess, it can rapidly consume all available $O_2$ and change to anaerobic metabolism producing an anaerobic environment and thus allow the anaerobic bacteria that are present to grow and cause disease. **Example:** *Escherichia coli* and *Salmonella.*

**Capnophilic Bacteria**

Bacteria which require $CO_2$ for their growth are called as capnophilic bacteria. **Example:** *Campylobacter.*

#### *Nutrition*

On the basis of their mode of nutrition bacteria are classified into two types namely autotrophs and heterotrophs.

**I Autotrophic Bacteria**

Bacteria which can synthesise their own food are called autotrophic bacteria. They may be further subdivided as

**A. Photoautotrophic bacteria**

Bacteria use sunlight as their source of energy to synthesize food. They may be

**1. Photolithotrophs**

In photolithotrophs the hydrogen donor is an inorganic substance.

**a. Green sulphur bacteria:** In this type of bacteria the hydrogen donor is $H_2S$ and possess pigment called **Bacterioviridin.** Example: *Chlorobium.*

**b. Purple sulphur bacteria:** For bacteria belong to this group the hydrogen donor is thiosulphate, **Bacteriochlorophyll** is present**.** Chlorophyll containing chlorosomes are present Example: *Chromatium*.

**2. Photoorganotrophs**

They utilize organic acid or alcohol as hydrogen donor. Example: Purple non sulphur bacteria – *Rhodospirillum.*

**B. Chemoautotrophic bacteria**

They do not have photosynthetic pigment hence they cannot use sunlight energy. This type of bacteria obtain energy from organic or inorganic substance.

**1. Chemolithotrophs**

This type of bacteria oxidize inorganic compound to release energy.

Examples:

1. Sulphur bacteria - *Thiobacillus thiooxidans*
2. Iron bacteria - *Ferrobacillus ferrooxidans*
3. Hydrogen bacteria - *Hydrogenomonas*
4. Nitrifying bacteria - *Nitrosomonas* and *Nitrobacter*



XIth_BIO-BOTANY_Ch_1-EM.indd 14 5/17/2020 11:09:31 PM